



Extensive variation within the pan-genome of cultivated and wild sorghum

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Sorghum is a drought-tolerant staple crop for half a billion people in Africa and Asia, an important source of animal feed throughout the world and a biofuel feedstock of growing importance. Cultivated sorghum and its inter-fertile wild relatives constitute the primary gene pool for sorghum. Understanding and characterizing the diversity within this valuable resource is fundamental for its effective utilization in crop improvement. Here, we report analysis of a sorghum pan-genome to explore genetic diversity within the sorghum primary gene pool. We assembled 13 genomes representing cultivated sorghum and its wild relatives, and integrated them with 3 other published genomes to generate a pan-genome of 44,079 gene families with 222.6 Mb of new sequence identified. The pan-genome displays substantial gene-content variation, with 64% of gene families showing presence/absence variation among genomes. Comparisons between core genes and dispensable genes suggest that dispensable genes are important for sorghum adaptation. Extensive genetic variation was uncovered within the pan-genome, and the distribution of these variations was influenced by variation of recombination rate and transposable element content across the genome. We identified presence/absence variants that were under selection during sorghum domestication and improvement, and demonstrated that such variation had important phenotypic outcomes that could contribute to crop improvement. The constructed sorghum pan-genome represents an important resource for sorghum improvement and gene discovery.

The rapidly changing global climate is compromising the capacity of agriculturalists to generate the productivity gains required to meet the growing demands for food, feed and fuel¹. It seems increasingly likely that producing new genetic variability through recombination within elite breeding populations will be insufficient to meet the challenges created by changing environments. Instead, the search for new variations will need to expand to encompass the entire species, including landraces and wild relatives. Sorghum, an adaptable stress-tolerant staple food crop for half a billion people in Africa and Asia, is grown in challenging environments that are at the forefront of agriculture's battle against climate change. Represented by diverse racial types displaying distinct morphological characteristics, cultivated sorghum has a complex domestication history. It is believed to have been first domesticated from its wild progenitor (*Sorghum bicolor* ssp. *verticilliflorum*) in Ethiopia and Sudan more than 5,000 years ago^{2,3}, with a second, more recent, domestication event in West Africa^{4,5}. The domestication and subsequent improvement of cultivated sorghum, which focus on selecting materials adapted to particular agro-ecological conditions and end uses, has led to a marked reduction of diversity in modern germplasms compared with their wild counterparts^{4,6,7}. Thus, utilization of broad genetic diversity from landraces and wild relatives in the cultivated gene pool is a critical element in sorghum improvement programmes, which requires characterizing and understanding genetic diversity within the primary gene pool as an entry point to its effective utilization.

To explore the diversity within the sorghum primary gene pool, 13 sorghum accessions including 1 *Sorghum propinquum*⁸, 3 *S. bicolor* ssp. *verticilliflorum*⁹, 1 *S. bicolor* ssp. *drummondii*⁹ and 8 *S. bicolor* ssp. *bicolor*⁹ were selected for de novo assembly to characterize the pan-genome (Supplementary Table 1 and Supplementary Fig. 1). The selected *S. bicolor* ssp. *bicolor* lines cover four diverse racial groups within cultivated sorghum in addition to two guinea-margaritifera lines, representing the second domestication in West Africa^{4,5}. All 13 genomes were sequenced using a combination of Illumina short reads and PacBio long reads. On average, we generated approximately 127 Gb of Illumina reads per genome, using a range of libraries and about 32.6 Gb of PacBio reads, including >60 Gb PacBio reads generated for sorghum accessions J12731 and 353 (Supplementary Tables 2 and 3). Using these sequencing data, the 13 genomes were assembled with N50 contig lengths of up to 3.5 Mb (Table 1, Supplementary Table 4 and Supplementary Fig. 2).

A total of 1.5 billion Hi-C read pairs were generated for four genomes (Supplementary Table 5), which validated more than 85% of the contig joins. We further used these data to correct misjoins in these assemblies and cluster and orient scaffolds, which placed more than 98% of the assembled genomes into chromosomes in 353 and J12731 (Supplementary Fig. 3). We also ordered and oriented scaffolds of the other nine genomes into chromosomes based on their synteny to the reference-quality genomes. As a result, the majority (averaging 90%) of the 13 genomes were placed into chromosomes, enabling differentiation between allelic and isoform

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Table 1 | Summary statistics of the 13 assembled sorghum genomes

Sample	Race	Assembled genome size	Scaffold N50 (bp)	Contig N50 (bp)	Annotated genes	Completeness (BUSCO)
S369-1	<i>S. propinquum</i>	549,286,155	327,824	288,992	32,176	93.5%
PI536008	<i>S. bicolor</i> ssp. <i>verticilliflorum</i>	672,504,365	332,628	185,088	31,898	95.2%
353	<i>S. bicolor</i> ssp. <i>verticilliflorum</i>	737,498,818	71,609,523	2,065,169	37,512	98%
AusTRCF317961	<i>S. bicolor</i> ssp. <i>verticilliflorum</i>	643,378,005	194,000	87,129	32,231	93.5%
PI532566	<i>S. bicolor</i> ssp. <i>drummondii</i>	643,043,272	173,705	75,814	32,457	93.4%
IS929	<i>S. bicolor</i> ssp. <i>bicolor</i>	579,961,993	356,166	342,003	36,193	97.0%
IS3614-3	<i>S. bicolor</i> ssp. <i>bicolor</i>	630,994,543	489,602	253,518	34,082	95.0%
IS19953	<i>S. bicolor</i> ssp. <i>bicolor</i>	600,155,976	396,104	385,210	35,352	97.4%
IS8525	<i>S. bicolor</i> ssp. <i>bicolor</i>	569,046,262	418,749	402,331	35,745	97.2%
PI525695	<i>S. bicolor</i> ssp. <i>bicolor</i>	468,122,292	153,289	139,434	33,295	97.2%
IS12661	<i>S. bicolor</i> ssp. <i>bicolor</i>	622,136,837	308,969	163,708	34,513	95.3%
R931945-2-2	<i>S. bicolor</i> ssp. <i>bicolor</i>	599,187,482	281,851	116,484	32,231	93.4%
Ji2731	<i>S. bicolor</i> ssp. <i>bicolor</i>	703,754,232	64,965,820	3,479,156	36,952	98.0%

variation and gene-content variation (Supplementary Table 6). The two most complete assemblies (353 and Ji2731) achieved a final contig N50 of greater than 2 Mb with 98% of the genome placed into chromosomes. The assemblies contained an average of 96.1% of the 1,440 benchmarking universal single-copy orthologues (BUSCO)¹⁰ (Supplementary Table 7).

We then predicted gene models from the 13 assemblies together with 3 publicly available assemblies (BTx623^{11,12}, Tx430¹³ and Rio¹⁴) using a strategy combining multiple ab initio gene predictions and evidence-based searches. The number of predicted genes per genome varied between 31,898 and 37,512 (Supplementary Table 8), with an average coding length of 1,072 bp and an average of 4.2 exons per gene. A maximum BUSCO value of 93.8% was achieved for predicted genes, with an average of 91.2% (Supplementary Table 7). More than 85% of the annotated genes are supported by homology to known proteins in other species. Around two-thirds of the predicted gene models were validated by the RNA-sequencing (RNA-seq) data generated from six selected lines (Supplementary Table 9).

Characterization of the sorghum pan-genome

Collectively, these 16 genomes captured the majority (>90%) of unmapped reads within a previous sorghum resequencing study⁴ when aligned against the reference genome (BTx623), resulting in an average of less than 0.4% unmapped reads per sample (Supplementary Table 10 and Supplementary Fig. 4). To estimate the pan-genome size, we identified sequences longer than 500 bp from each genome assembly that were absent from the BTx623 genome (<25% coverage). After filtering out redundant sequences, we obtained 244,825 sequences with a N50 of 2,000 bp, totalling 222.6 Mb (Supplementary Table 11). Repetitive elements accounted for 39% of the non-reference genome, which was less than the value for the 16 genomes. Combining these newly identified sequences with the reference genome provided a sequence-based pan-genome of 954.8 Mb, 30% larger than the 732.2 Mb reference genome (Supplementary Fig. 5). Sixty-two percent of the pan-genome sequence was shared by all 16 genomes. The large amount of new sequence identified highlights the inadequacy of a single reference genome to represent the diversity of a species. A graph-based pan-genome was generated using minigraph¹⁵ (Extended Data Fig. 1).

We clustered the predicted genes from all 16 genomes into gene families on the basis of sequence similarity and further checked for the presence of the identified gene families and singleton genes across the 16 genomes using a BLAST approach, leading to the

identification of 44,079 gene families in total. We iteratively selected lines from the 16 genomes to simulate sorghum pan-genome expansion, which showed that the addition of new genes decreases as the number of genomes increases, in line with other plant species^{16–18} (Supplementary Fig. 6). The 44,079 gene families were further categorized on the basis of their presence in each genome. There 15,867 (36%) gene families were present in at least 15 genomes, defined as core; 28,026 (63.6%) gene families were present in 2–14 genomes, defined as shell; and 186 (0.4%) gene families were present in only one genome, defined as cloud (Fig. 1a,b). The percentage of core gene families was lower than that reported in rice¹⁸ (46%), soybean¹⁶ (51%) and *Brachypodium distachyon*¹⁷ (55%). There were 18,481 singleton genes that were found exclusively in individual sorghum accessions. On average, each individual sorghum genome consisted of 58.8% core genes, 37.9% shell genes and 3.3% private genes (Fig. 1c). In general, the wild relatives of sorghum had larger numbers of private genes, with *S. propinquum* being the largest single contributor (Supplementary Fig. 7). Around half of the dispensable gene families (shell and cloud genes) were not present in the reference genome (BTx623), and these absences were primarily (62%) due to presence/absence variation.

We selected a representative from each gene family to further investigate the gene-based pan-genome. Compared with core genes, shell and cloud genes were found to be shorter in length and with fewer exons (Extended Data Fig. 2). However, shell genes exhibited a higher nucleotide diversity (π) and a higher non-synonymous/synonymous substitution ratios (K_a/K_s) than core genes, indicating that they evolve faster and are less functionally conserved (Fig. 1d,e). Expression analysis using our RNA-seq data showed that a much higher percentage of core genes (averaging 83%) were expressed compared to shell (averaging 61%) and cloud genes (averaging 47%) with a significantly higher level of expression ($P < 0.01$, two-sided Wilcoxon signed-rank test; Extended Data Fig. 3 and Supplementary Table 12), consistent with reports in soybean¹⁶, *B. distachyon*¹⁷, rice¹⁹ and tomato²⁰. Functional enrichment analysis identified that dispensable genes were significantly enriched in biological process pathways, such as secondary metabolic processes, cellular amide metabolic processes and amino acid transport (Supplementary Fig. 8). By contrast, core genes were enriched for basic, critical functions including developmental process, RNA processing, reproductive system development, leaf development, seed development, cell differentiation and chloroplast organization (Supplementary Fig. 9).

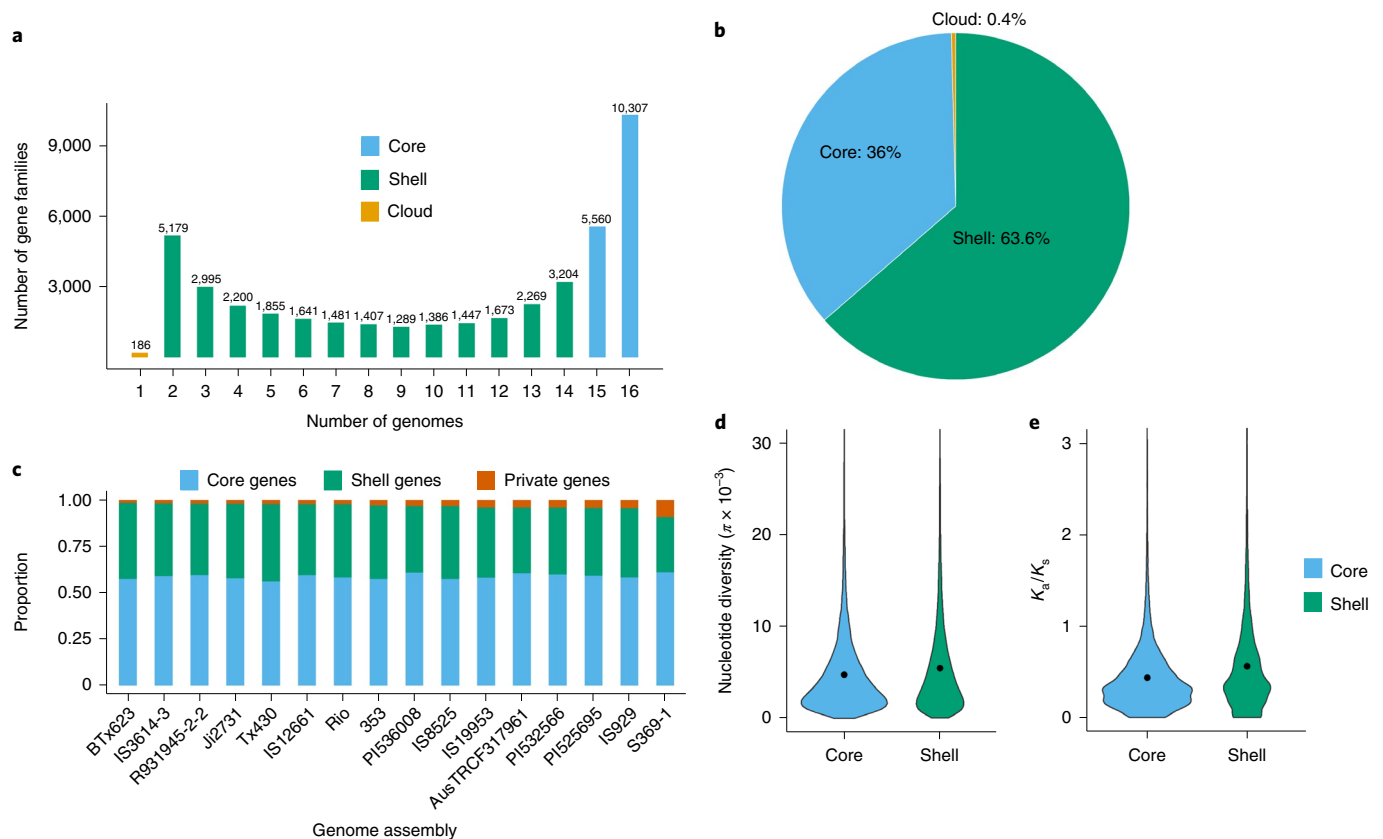


Fig. 1 | Sorghum pan-genome. The gene-based sorghum pan-genome was created by clustering annotated genes from 16 genomes into gene families. **a**, Number of gene families represented versus number of genomes. **b**, Composition of core, shell and cloud genes in the sorghum pan-genome. **c**, Composition of core, shell and private genes per genome. **d**, Violin plots showing nucleotide diversity of core genes and shell genes. Shell genes are more diverse than core genes ($P = 8.2 \times 10^{-15}$, two-sided t -test). **e**, Violin plots showing K_a/K_s values of core and shell genes. Shell genes have higher K_a/K_s values compared with core genes ($P < 2.2 \times 10^{-16}$, two-sided Wilcoxon signed rank). Black dots indicate mean values.

Extensive genetic variations within sorghum pan-genome

To identify sequence variations within the sorghum pan-genome, the other 15 genome assemblies were anchored to the reference genome (BTx623). In total, 15,293,465 single nucleotide polymorphisms (SNPs) were identified, with an average of 2.2 million (ranging from 1.2 million to 5.3 million) SNPs per sample (Supplementary Table 13). Among these SNPs, 406,444 were non-synonymous, including 1,913 causing splice changes, 4,904 causing premature stop codon changes and 1,960 changes to stop or start codons based on the annotation of BTx623. Each gene contained, on average, 11.9 missense SNP sites including 0.3 large-effect SNPs (SNPs causing changes of start or stop codons and splice sites). We constructed phylogenetic relationships among the 16 genomes using the SNP data, which showed separate lineages for guinea-margaritiflorum and cultivated sorghums derived from *S. bicolor* ssp *verticilliflorum* (Fig. 2a), supporting previous reports of a second independent domestication event in West Africa^{5,21}.

In addition to SNPs, between 0.3 and 1.5 million insertions/deletions (indels) were identified per genome following the pairwise comparison with the reference genome (BTx623), more than 80% of which were less than 6 bp in size (Supplementary Table 14). Around two-thirds of the indels were located in intergenic regions, and between 9,646 and 81,489 indels were located in coding regions, with around half of them causing frameshifts. Compared with the BTx623 genome, we also identified between 429 and 1,118 genes in the other 15 genomes exhibiting copy-number variation (CNV) (Supplementary Table 15). Gene Ontology (GO) analysis indicated that biological processes such as defence response, signal

transduction and cellular protein catabolic process were enriched in CNV genes (Supplementary Fig. 10).

We identified between 19,359 and 147,899 presence/absence variants (PAVs) in each comparison, with the total size varying between 13.3 Mb and 102.4 Mb (Supplementary Table 16). As the most phylogenetically distant genome from the reference (BTx623) in this study, *S. propinquum* was found to have the largest amount of SNPs, indels, CNVs and PAVs when compared with the reference. We constructed the phylogenetic relationships among the 16 genomes using the identified PAVs and found that it largely resembled the phylogeny based on SNPs, validating the reliability of the PAV detection approach (Supplementary Fig. 11). There were a total of 101,839 reference-present PAVs (PAVs present in BTx623), totalling 64.3 Mb. These reference-present PAVs overlapped with 12,542 previously predicted gene models¹², and an additional 9,072 gene models contained PAVs in the 2 kb promoter region, highlighting possible widespread effects of PAV on gene functionality. There were 2,514 reference genes severely affected by PAVs (>50% of coding region covered by PAVs) (Supplementary Table 17), which were enriched for biological processes related to biotic and abiotic stress, such as defence response, cell redox homeostasis, signal transduction, cell communication and secondary metabolite biosynthetic process (Supplementary Fig. 12). In particular, disease resistance genes were significantly enriched in this set of genes (Supplementary Table 18; $P < 0.01$, Fisher's exact test). Furthermore, some of these PAV-affected genes have key roles in genetic control of important agronomical traits, such as *LGS1* (Sobic.005G213600), which encodes a sulfotransferase and whose deletion confers

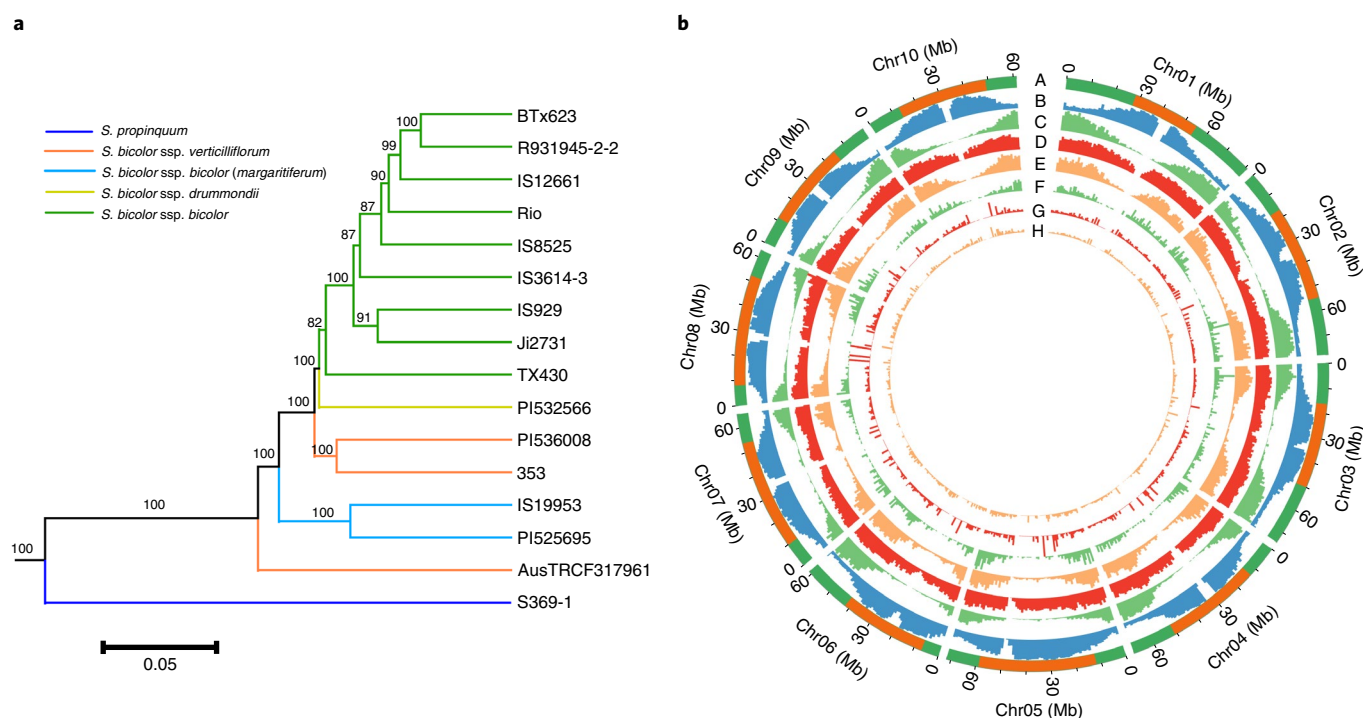


Fig. 2 | Phylogenetic relationships and distribution of genetic variation across the sorghum genome. a, Phylogenetic relationships based on SNPs among 16 genomes in the sorghum pan-genome. The numbers next to the nodes represent bootstrap values. **b**, Distribution of genetic variation across the sorghum genome. Lanes: A, chromosome ideograms for *S. bicolor*, showing pericentromeric (orange) and telomeric (green) regions; B, TE density; C, gene density; D, SNP density; E, PAV density; F, CNV density; G, proportion of genes affected by large-effect mutations; H, proportion of genes affected by PAVs. In lanes G and H, proportions were calculated using a 1-Mb window.

reduced germination stimulation of the parasitic weed *Striga*²², *Dry* (Sobic.006G147400), which encodes a plant-specific NAC transcription factor whose absence results in juicy sorghum stem²³.

These sequence variations were distributed unevenly throughout the genome. We observed a positive correlation between the distributions of PAVs and SNPs ($r^2=0.7$; Supplementary Fig. 13). We also observed a gradual increase of SNP density within 2 kb of PAV events (Supplementary Fig. 14), supporting the structural-variation-associated mutation hypothesis^{24,25}. Consistent with reports in soybean¹⁶ and *Zea mays*²⁶, genes in telomeric regions, where recombination is increased, were less likely to be affected by large-effect mutations (including SNPs or indels causing changes of start/stop codons, frameshifts and splice site mutations) and PAVs compared with pericentromeric regions (Fig. 2b). We also found that transposable elements (TEs), particularly *hAT* transposons, *Helitron* and long interspersed nuclear elements, which have been associated with mediating sequence movement (Supplementary Table 19), were enriched in the 2-kb vicinities of PAVs, indicating that TEs may have a role in generating PAVs.

Selection of PAVs during sorghum domestication

To investigate the roles of PAVs during sorghum domestication, we conducted a fixation index (F_{ST}) analysis of 346,664 PAVs genotyped across a panel of 44 sorghum accessions including wild sorghum, landraces and improved inbred lines⁴. In total, we identified over 4,000 PAVs under selection in pairwise comparisons between wild lines, landraces and improved lines (Supplementary Table 20). The selection signals identified using PAVs were largely consistent with those detected using SNPs, with around 75% of PAVs under selection co-locating with SNPs under selection (Supplementary Figs. 15–17). Although PAVs under selection during domestication contained approximately the same number of genes (595) as PAVs

under selection during improvement (641), the number of genes that were lost or gained during domestication (92 out of 120) was far greater than those lost or gained during improvement (15 out of 20) (Supplementary Tables 21–23). The genes affected by the PAVs under selection were enriched for biological processing, oxidation–reduction processing, protein phosphorylation and regulation of actin filament polymerization (Supplementary Fig. 18).

A number of the PAVs identified as being under selection contained genes that are related to the domestication syndrome, including loss of shattering, loss of dormancy and increase in grain size. Previous research has identified three functional mutations of the *Shattering1* gene, including a 2.2-kb deletion underlying the loss of seed shattering during sorghum domestication²⁷. Our analysis showed that this 2.2-kb deletion in *Shattering1*, which was absent in wild lines but present in 80% of improved lines, was under positive selection in the comparison between wild and improved lines. We also found a 243-bp deletion between the third and fourth exons of *Sobic.006G141700*, the orthologue of *OsGrxC2.2* regulating grain weight in rice²⁸, under selection during sorghum domestication. *Sobic.006G141700* co-locates with a previously reported grain weight QTL identified between cultivated sorghum and its wild progenitor²⁹. Additionally, a 5-kb deletion containing *Sobic.009G114300*, the orthologue of *PHR1* regulating seed discoloration in rice³⁰, was identified as being under selection during sorghum domestication. *PHR1* was reported being under contrast selection in *japonica* and *indica* rice for aesthetic appeal and possible relation to seed dormancy, respectively³⁰.

PAVs underlie grain-colour variation in sorghum

To further test the impact of PAVs on agronomic traits in sorghum, we conducted a genome-wide association study (GWAS) on

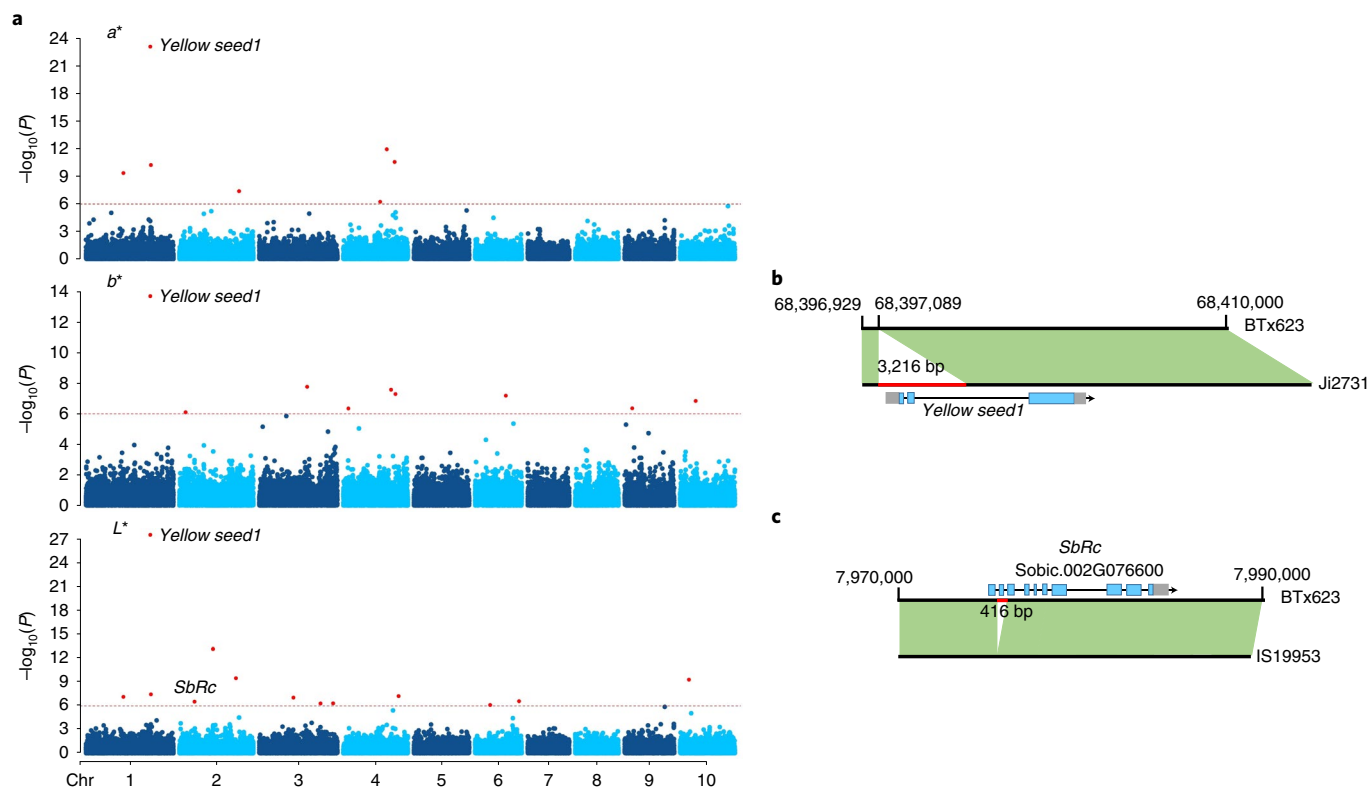


Fig. 3 | Presence/absence variation underlying grain-colour variation in sorghum. a, Manhattan plots for genome-wide association mapping of 3 grain-colour coordinates (lightness, L^* ; red/green value, a^* ; and blue/yellow value, b^*) in a diversity panel of 839 accessions. The red dashed lines indicate the significance threshold. **b,c**, Graphic representation of sequence differences in the *Yellow seed1* (**b**) and *SbRc* (**c**) loci. Sequence coordinates are based on the sorghum reference genome (v.3.1.1). Sequence displaying presence/absence variation is marked in red. Gene structures are represented with solid lines (gene space), blue rectangles (exon) and grey rectangles (untranslated region). Arrows indicate gene direction.

grain colour using a previously reported diversity panel of 839 individuals³¹. Grain colour, measured as lightness (L^*), red/green value (a^*) and blue/yellow value (b^*) coordinates was assessed in two field trials and showed extensive variation with high heritability (Supplementary Fig. 19 and Supplementary Table 24). GWAS analysis of the 3 coordinates identified 25 SNPs significantly associated with grain colour (Fig. 3a, Supplementary Fig. 20 and Supplementary Table 25), co-locating with 5 previously reported candidate genes for grain colour in sorghum, rice and maize.

The most significant SNP across all colour coordinates (chromosome 1: 68372486) corresponded with *Yellow seed1*, a sorghum MYB gene controlling phlophenes accumulation in pericarp³². Sequence analysis of *Yellow seed1* in the sorghum pan-genome identified a PAV of 3,216 bp, which showed significant association with variation of all three grain-colour parameters (Supplementary Fig. 21). Absence of this 3.2-kb sequence resulted in the deletion of the 5' untranslated region and first two exons of *Yellow seed1* in 4 sorghum lines, including the reference genome (BTx623) (Fig. 3b). Additionally, *SbRc* (Sobic.002G076600), the candidate gene for a SNP significantly associated with L^* on chromosome 2: 7828095 was identified as the orthologue of *Rc*, which controls grain colour in rice³³ (Fig. 3c). Sequence analysis of *SbRc* in the sorghum pan-genome identified a 416-bp PAV, which led to deletion of the second exon in 3 sorghum accessions. Further testing showed that this PAV was significantly associated with the grain-colour parameters, a^* and L^* (Supplementary Fig. 21). These results demonstrate the impact of PAVs on variation of grain colour in sorghum.

Discussion

Understanding the genetic diversity ranging from single nucleotide variation to large-scale structural variation within crop species and their association with agronomic traits is critical to crop improvement. Over the past two decades, advances in sequencing technologies have enabled resequencing studies to explore sequence variation within crop species and identify genomic regions associated with domestication and adaptation traits^{4,34,35}. However, resequencing studies, which identify genomic variation through mapping of short reads to a single reference genome, preclude the discovery of regions with marked structural variability, and are therefore unable to fully capture structural and gene-content variation. Most recently, whole-genome comparisons within a species through the development of pan-genomes have revealed substantial gene-content and structural variation, leading to the realization that a single reference genome is inadequate to capture the full landscape of diversity within a species^{16,17,19,36–39}. This study presents the first pan-genome developed for sorghum based on 16 genomes including representatives of diverse cultivated races and inter-fertile wild relatives. It provides a key resource for increasing productivity and stress tolerance of this globally important food crop. The 13 de novo assemblies developed in this study, together with assemblies of other key sorghum lines^{12–14,40}, provide a solid foundation for understanding trait biology and mining genetic diversity. The constructed sorghum pan-genome showed extensive gene-content variation, with 64% of gene families showing variable existence among genomes. The proportion of dispensable genes in the sorghum pan-genome is higher than that reported in rice¹⁸ (54%), soybean¹⁶ (49%) and *B. distachyon*¹⁷ (45%), which were

also constructed using the de novo assembly approach. As a genetic property of a species, the proportion of dispensable genes in the pan-genome is likely to be affected by multiple factors, including the ploidy level, outcrossing rate, genome size and TE content³⁷. The high proportion of dispensable genes in our sorghum pan-genome is probably due to a combination of higher outcrossing rate and TE content compared with rice, soybean and *B. distachyon*. It is also probably a reflection of the wide genetic representation encompassed by our sorghum pan-genome, which includes inter-fertile wild genotypes. The comparison between core genes and dispensable genes suggested that dispensable genes might have an important role in adaptation to abiotic and biotic stresses. The 222.6 Mb of new sequence absent from the reference genome (BTx623), together with the wide range of genetic variation uncovered by whole-genome comparisons represents a valuable resource that has not been previously accessible to the research and breeding community relying on the 'BTx623' reference genome. The identified structural variation underlying variation of grain colour highlights the value of the new variation uncovered in the sorghum pan-genome for crop improvement and biological discoveries. Overall, this sorghum pan-genome characterizes new genetic diversity within the primary gene pool, which will be critical for meeting the challenges of improving productivity in the light of a growing world population and climate change.

Methods

Sequencing of 13 sorghum genomes. Thirteen sorghum accessions were selected for de novo assembly to represent the major racial groups and geographical distribution of the sorghum primary gene pool (Supplementary Table 1). High-molecular-mass nuclear DNA isolated from young leaf tissue was used to construct Illumina sequencing libraries (including short-insert libraries (180 bp and 500 bp) and mate-pair large-insert libraries (2 kbp and 5 kbp) and PacBio SMRT libraries (20 kbp) for each sample, following standard protocols. The libraries were sequenced on the Illumina HiSeq 4000 platform and the PacBio Sequel platform at Beijing Genomic Institute (Shenzhen, China) and BioMarker Technologies.

Raw reads generated from the Illumina platform were cleaned before subsequent analyses, to filter out low-quality sequence, according to the following criteria: (1) putative PCR duplicates generated by PCR amplification in the library construction process; (2) reads with >10 nt aligned to the adaptor; (3) reads with ≥20% low-quality bases that having phred quality <7; (4) Reads with ≥4% N bases.

The PacBio reads were corrected using filtered Illumina short reads with proovread (v2.14)⁴¹.

HindIII-digested Hi-C libraries were constructed using high-quality DNA extracted from young leaves following the proximo Hi-C protocol. Sequencing of Hi-C libraries was performed on the Illumina NovaSeq 6000 platform at BioMarker Technologies. Raw Hi-C data was cleaned by removing contamination, adaptor sequence and low-quality data using SOAPnuke (v1.5.6)⁴².

De novo assembly. The genome size of each sorghum accession was estimated via generating a 17-mer depth distribution of short-insert paired-end reads using Jellyfish⁴³. Two de novo assembly strategies were used in this study according to data availability. For the two accessions (353 and Ji2731) with >80X PacBio data, PacBio reads were firstly corrected using Canu (v1.7.1)⁴⁴, which were then assembled using WTDBG⁴⁵. Contigs were polished using Illumina short reads with Pilon⁴⁶.

For the remaining accessions, a hybrid assembly strategy similar to one previously reported⁴⁷ was used. Specifically, reads of 180-bp and 500-bp libraries were firstly used to build contigs with Platanus (v1.2.4)⁴⁸ with parameter '-k 65 -c 3 -t 80 -m 500 -u 0.2'. The resultant contigs were then combined with corrected PacBio reads for hybrid assembling using DBG2OLC⁴⁹ with parameter 'k 17 LD1 0 KmerCovTh 3 MinOverlap 20 AdaptiveTh 0.005 MinLen 50 RemoveChimera 1'. Assembled hybrid contigs were further corrected using Illumina short reads with Pilon⁴⁶. These 2-kbp and 5-kbp meta-pair reads were used for scaffolding corrected hybrid contigs with SSPACE (v2.0)⁵⁰ with parameter '-x 0 -k 3 -a 0.8 -n 15 -z 0 -T 8'. During the scaffolding process, only mate-pair reads with unique match positions (coverage > 90%) were used to avoid false joints. Finally, assembled scaffold sequences were BLASTed against bacterial genomes, and possible contamination was excluded (identity ≥ 95% and e-value ≤ 10⁻⁵).

HiC-Pro (version 2.8.1)⁵¹ was used to extract the valid interaction information from clean Hi-C reads that were trimmed at putative Hi-C junctions, as follows: (1) reads were aligned to the genome using BWA aligner (version 0.7.12)⁵² with default parameters and uniquely aligned reads were extracted; (2) valid interaction

reads were extracted; (3) invalid read pairs, including dangling-end and self-cycle, re-ligation and dumped products were removed. The interaction reads were then aligned to scaffolds that were cut into 300-kb windows and information from the resultant alignment was used to assess and correct contig joins in our assemblies and join the scaffolds using LACHESIS⁵³.

Gene prediction and annotation. Gene models were predicted using an approach integrating ab initio predictions, homology-based prediction and RNA-seq. For ab initio prediction, Augustus (version 2.5.5)⁵⁴ and SNAP⁵⁵, were used to predict gene models. For homology-based prediction, protein sequences of *S. bicolor* (v3.1.1) and 4 closely related species including *B. distachyon* (v3.1), *Setaria italica* (v2.2), *Z. mays* (v4) and *Oryza sativa* (v7) were used to search for putative gene coding sequences using MAKER⁵⁶. RNA-seq data of mixed tissues were mapped to each assembled genome using HISAT2 (v2.1.0) and transcriptome-based gene structure were obtained using StringTie (v1.3.4). The different sets of gene predictions for each genome were integrated using MAKER⁵⁶ to produce a consensus set of gene models for each genome.

Functions of predicted genes were inferred on the basis of their best BLASTP hits against SwissProt (<https://www.uniprot.org/>), TrEMBL (<https://www.uniprot.org/statistics/TrEMBL>) and NR databases. Motifs and domains were identified via searching protein databases including ProDom, PRINTS, Pfam, SMART, PANTHER and PROSITE using InterProScan (<http://www.ebi.ac.uk/interpro/>). GO terms for each gene were obtained by converting NR accession IDs to GO terms. KEGG pathway membership was determined by the predicted genes' best match when aligning against KEGG proteins (<https://www.genome.jp/kegg/>).

RNA-seq of six sorghum accessions. Six samples (IS3614, IS8525, PI525695, 273, S353 and IS929) were grown in a glass house at The Institute of Botany, Chinese Academy of Science (Beijing, China). For each sample, total RNAs from five different tissues (young leaf, root, stem, young inflorescence and immature seed) were extracted and mixed equally to construct sequencing libraries following standard protocols. These libraries were then sequenced on the BGISEQ-500 platform at the Beijing Genomic Institute (Shenzhen, China) with an average of 15.7 Gb clean data generated for each sample.

Pan-genome construction. To identify related genes across assemblies, OrthoMCL (v2.0.9)⁵⁷ was used to cluster predicted gene models of 16 genomes into gene families with default settings (BLASTP e-value < 10⁻⁵ and inflation factor = 1.5). Similar to the recently constructed maize pan-genome⁵⁸, sequences of all members with the identified gene families and singleton genes were aligned to genomes where these genes were not annotated using BLAST to recover potential missing genes in the annotation process. Genomes with sequence in the same syntenic region to an annotated gene and aligned to the annotated gene with greater than 90% coverage and greater than 90% identity was considered as having this gene. Synteny among the 16 genomes were obtained using MCSanX⁵⁹.

Identified gene families were then classified into core, shell and cloud on the basis of the number of sorghum accessions in which they were identified. Core gene families refers to gene families present in more than 15 accessions. Shell gene families refers to gene families present in more than one but less than 15 accessions. Cloud gene families refer to gene families only present in one accession. A graph-based pan-genome was constructed with the BTx623 genome as a reference using minigraph¹⁵.

Detection of SNPs and indels. To identify SNPs and indels (<100 bp) in the sorghum pan-genome, the reference genome (BTx623) was aligned to each of the other 15 genomes using the NUCmer program from the MUMmer package (v3.0)⁶⁰ with parameter '-maxmatch -c 90 -l 40'. SNPs and small indels (<100 bp) were extracted from one-to-one alignment regions filtered using the delta-filter program in the MUMmer package.

Identification of PAVs. PAVs were identified using an approach similar to one previously reported⁶¹. We compared each genome with the reference genome (BTx623) using a sliding-window method. In each comparison, to identify reference-absent PAVs, the non-reference genome was divided into 500-bp sliding windows with a step size of 100 bp. Each window was then aligned to the reference genome using BWA (v0.7.17)⁵² as follows: (1) The windows that could be aligned (>90% coverage) to the reference genome were combined; (2) the sequences of windows that could not be aligned to the reference genome with coverage ≥ 25% were combined; and (3) the identified unmapped regions of the non-reference genome in step 2 that overlapped with the mapped regions of the non-reference genome in step 1 were further excluded to identify the reference-absent PAVs.

Reference-present PAVs were identified as follows: (1) reference genome regions without a match (>90% coverage) to the non-reference genome windows identified in the previous steps were identified and combined; (2) contigs of the non-reference genomes were mapped onto the reference genome, and the gap regions on the reference genome that fell between matching regions of contigs were identified; (3) the unmatched regions of the reference genome identified in step 1 that overlapped with gap regions identified in step 2 were filtered out, resulting a final set of unmatched regions of the reference genome; (4) the reference

genome was divided into 500-bp sliding windows with a step size of 100 bp, and the windows were then aligned to the non-reference genome using BWA to identify those regions that could not be aligned to the non-reference genome with coverage $\geq 25\%$. Identified unmapped windows were combined, and those that overlapped with unmatched regions of the reference genome were identified as reference-present PAVs. (5) Illumina short reads of the non-reference genome were BLASTed against the identified reference-present PAVs in step 4 to exclude these that could be mapped with Illumina short reads. The collection of reference-absent PAVs and reference-present PAVs were identified for each genome separately in comparison to the reference.

Phylogenetic tree construction. Both the SNP and PAV datasets were used to calculate p-distance to determine the genetic distance among individuals. Based on genetic distance, phylogenetic trees were constructed using unweighted pair-group method with arithmetic mean (UPGMA) clustering algorithm in the Neighbour program of PHYLIP (v3.696) (<http://evolution.genetics.washington.edu/phyli.html>). The constructed tree file was visualized using Figtree (<http://tree.bio.ed.ac.uk/software/figtree/>).

GWAS analysis of grain colour. A previously reported diversity panel of 839 individuals³¹ was used in the GWAS analysis of grain colour. The population was planted in two field trials at Hermitage Research Facility and Gatton Research Facility, Queensland, Australia during the summer cropping seasons of 2015–2016. A row-column design with partial replication was applied in both trials. Grain colour was measured using SeedCount SC5000 (Next Instruments), which identifies colour difference using L^* , a^* and b^* coordinates. Each grain-colour coordinate was analysed as a multi-environment trial by fitting a linear mixed model using the package ASReml⁶² to calculate BLUPs. The resultant BLUPs were used as input phenotypes for GWAS analysis in FarmCPU⁶³ with 49,586 imputed SNPs (minor allele frequency > 0.01) from a previous study³¹. The first three principal components were used to control population structure. The Bonferroni correction was applied to define the threshold for significant association.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The datasets generated during and/or analysed during current study have been deposited in China National GeneBank database (<https://db.cngb.org>) under the project CNP0001440 and the Genome Sequence Archive⁶⁴ in the National Genomics Data Center⁶⁵, Beijing Institute of Genomics (China National Center for Bioinformation), Chinese Academy of Sciences under accession number CRA003806, which are publicly accessible at <https://bigd.big.ac.cn/gsa>.

Code availability

The code used in this manuscript is available at the GitHub repository https://github.com/xujiabao507/Sorghum_pangenome.

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Author contributions

E.M., H.J., D.J. and Y.T. designed this study and coordinated the project. Y.T., X.Z., A.C. and A.H. selected samples and conducted field work. T.S., Y.L. and X.W. collected samples. J.X. and F.T. carried out the genome assembly and annotation. Y.T., H.L. and F.T. performed pan-genome analysis. Y.T. and J.X. conducted variation detection, phylogenetic analysis and selection analysis. Y.T., X.Z. and A.H. carried out GWAS analysis. Y.T. wrote the manuscript, E.M. and D.J. edited the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41477-021-00925-x>.

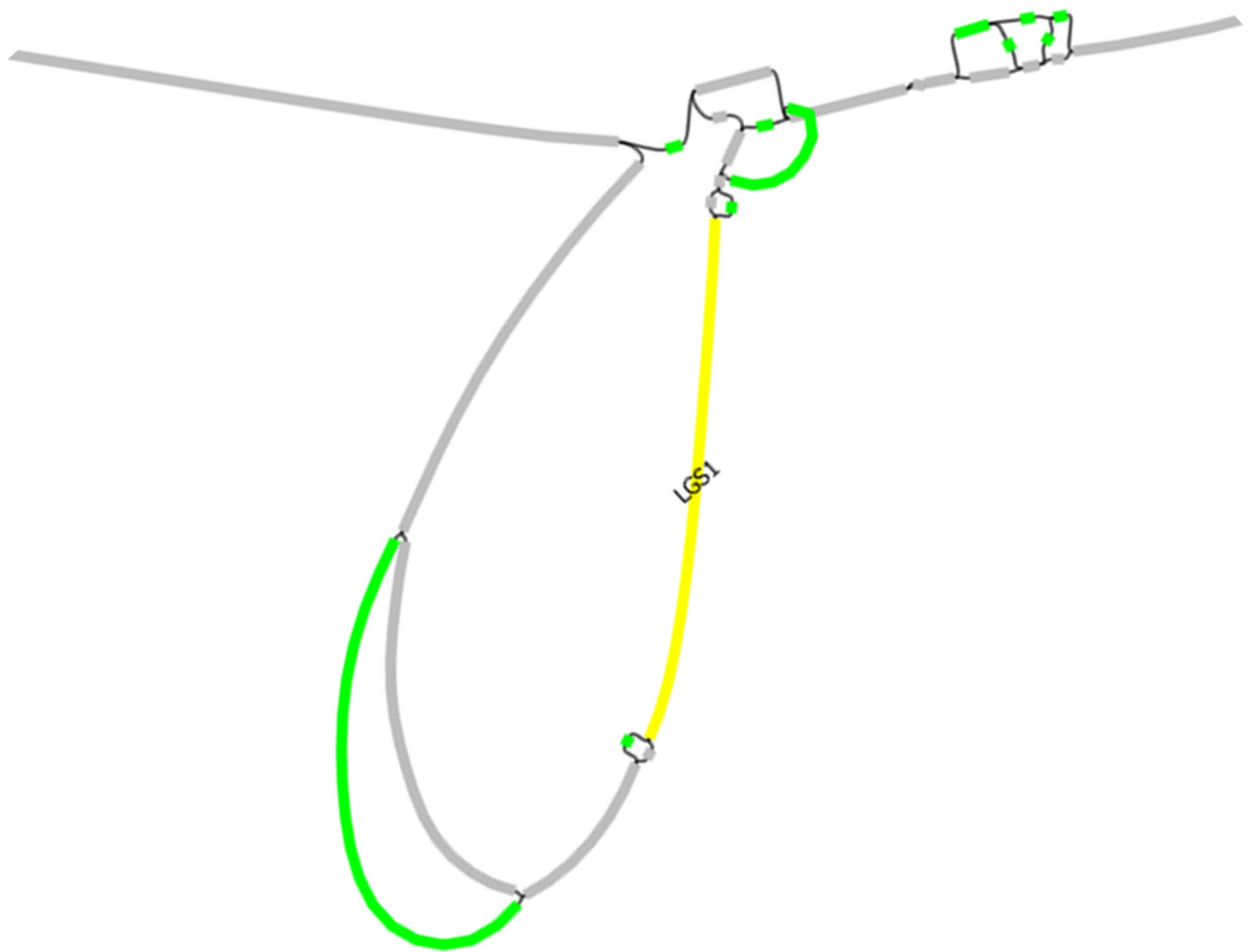
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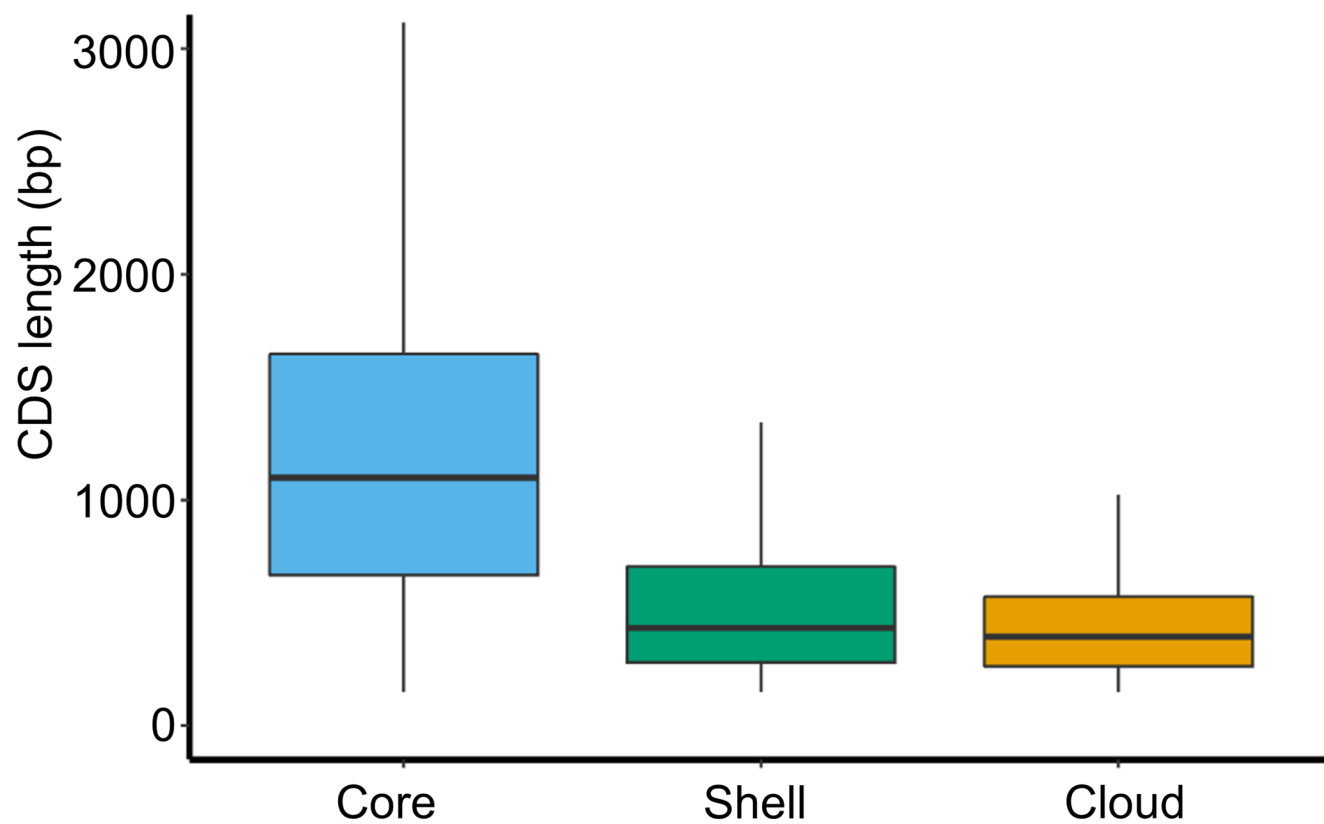
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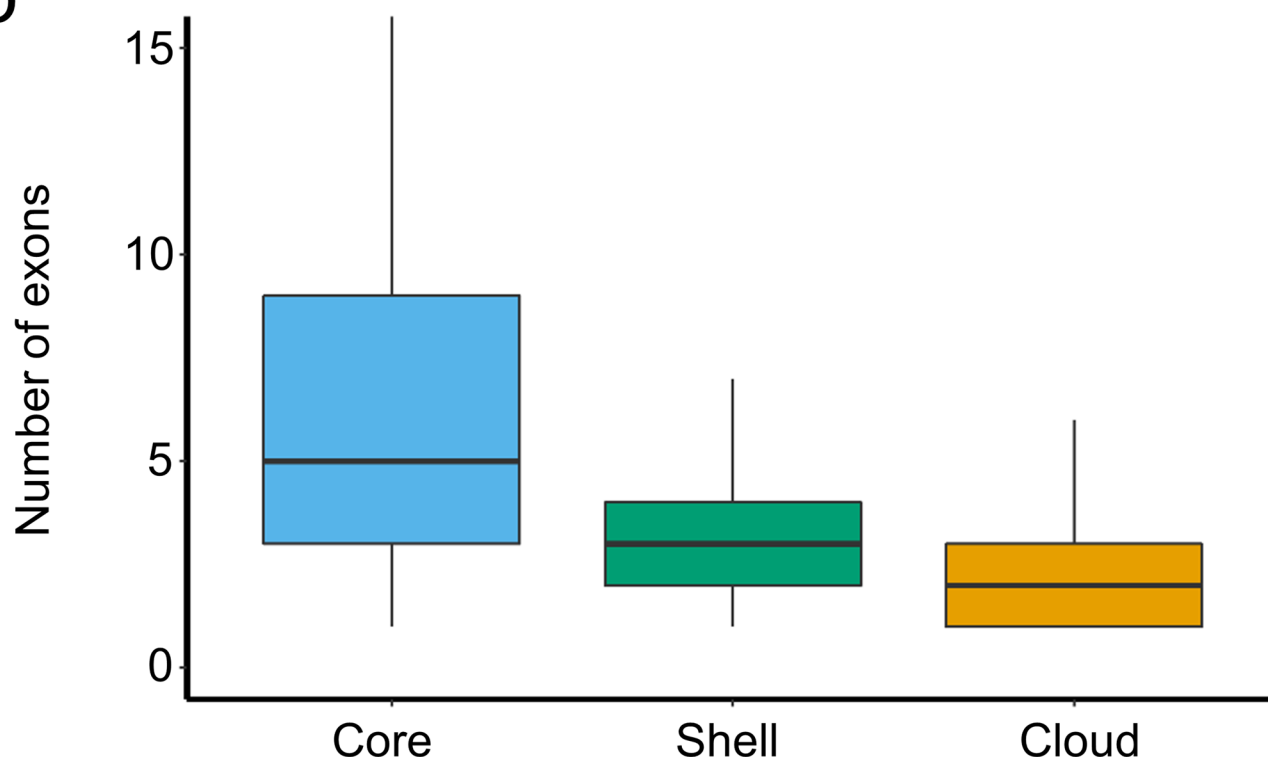


Extended Data Fig. 1 | A snapshot of graph-based sorghum pan-genome. This pan-genome graph shows variation within a *LGS1* region on Chromosome 5. The graph was visualised using Bandage. Yellow colour highlights the sequence segment containing *LGS1*. Grey colour indicates sequence segments from the reference genome BTx623. Green colour indicates sequence segments from genomes other than BTx623.

a

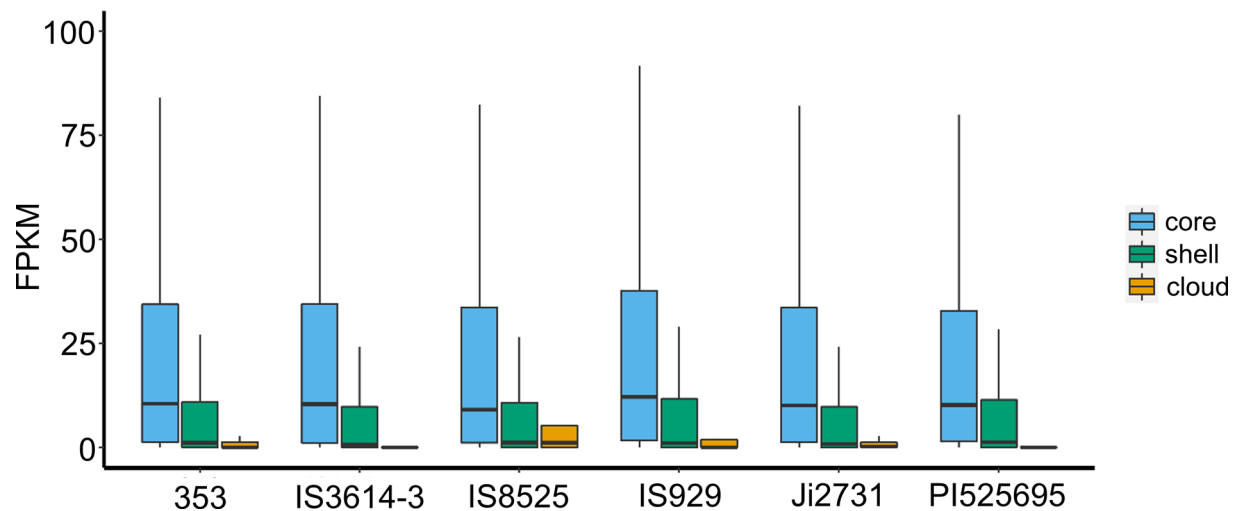


b



Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Comparison between core, shell and cloud genes. A CDS length. Core genes are significantly longer than shell and cloud genes ($p\text{-value} < 2.2 \times 10^{-16}$, Wilcoxon signed rank, two-sided). B number of exons. Core genes have significantly more exons than shell and cloud genes ($p\text{-value} < 2.2 \times 10^{-16}$, Wilcoxon signed rank, two-sided). Sample size: core, $n = 15,867$; shell, $n = 28,026$; cloud genes, $n = 186$. In the box plots, center lines represent the median, the bottom and top of boxes represent the first and third percentiles, whiskers show the data that lie within the 1.5 interquartile range of the first and third quartiles.



Extended Data Fig. 3 | Comparison of expression level between core, shell and cloud genes. Expression level (FPKM, Fragments Per Kilobase of transcript per Million mapped reads) of core, shell and cloud genes were measured in six samples. Core genes consistently showed a higher expression level compared to shell and cloud genes across six genomes ($p\text{-value} < 2.2 \times 10^{-16}$, Wilcoxon signed rank, two-sided). Sample size in the six genomes, 353: core, $n = 22,522$; shell, $n = 13,873$; cloud, $n = 78$, IS3614-3: core, $n = 20,786$; shell, $n = 12,648$; cloud, $n = 12$, IS8525: core, $n = 21,223$; shell, $n = 13,365$; cloud, $n = 12$, IS929: core, $n = 21,702$; shell, $n = 12,860$; cloud, $n = 35$, Ji2731: core, $n = 22,251$; shell, $n = 13,874$; cloud, $n = 57$, PI525695: core, $n = 20,445$; shell, $n = 11,372$; cloud, $n = 35$. In the box plots, center lines represent the median, the bottom and top of boxes represent the first and third percentiles, whiskers show the data that lie within the 1.5 interquartile range of the first and third quartiles.

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Software and code

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Data collection no software was used

Data analysis The code used in this manuscript is available at the GitHub repository: https://github.com/xujiabao507/Sorghum_pangenome. De novo assembly in this study was conducted using Canu(v1.7.1) Platanus (v1.2.4), DBG2OLC and SSPACE(v2.0). HiC-Pro (version 2.8.1) was used to analyze Hi-C data. MUMmer package (v3.0) was used to identify SNP and indels from genome comparison. BWA (v0.7.17) was used to align genome sequence. the unweighted pair-group method with arithmetic mean (UPGMA) clustering algorithm in PHYLIP (v3.696) was used to construct phylogenetic trees. OrthoMCL (v2.0.9)57 was used to cluster predicted genes into gene families. HISAT2 (v2.1.0) and StringTie (v1.3.4) were used to analyze RNA-seq data. All softwares used in this study have been disclosed in the method section.

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The datasets generated during and/or analysed during current study have been deposited in China National GeneBank database (<https://db.cngb.org>) under the

project CNP0001440 and the Genome Sequence Archive⁶³ in National Genomics Data Center⁶⁴, Beijing Institute of Genomics (China National Center for Bioinformation), Chinese Academy of Sciences under accession number CRA003806 that are publically accessible at <https://bigd.big.ac.cn/gsa>. Public database SwissProt (<https://www.uniprot.org/>), TrEMBL (<https://www.uniprot.org/statistics/TrEMBL>) and InterProScan (<http://www.ebi.ac.uk/interpro/>) were used to predict functions of predicted genes.

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Sample size	16, 16 representative genomes were selected in this study to capture the diversity within sorghum and balance the cost of sequencing
Data exclusions	no
Replication	these samples were selected for DNA sequencing, replication is not needed for this purpose.
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☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging